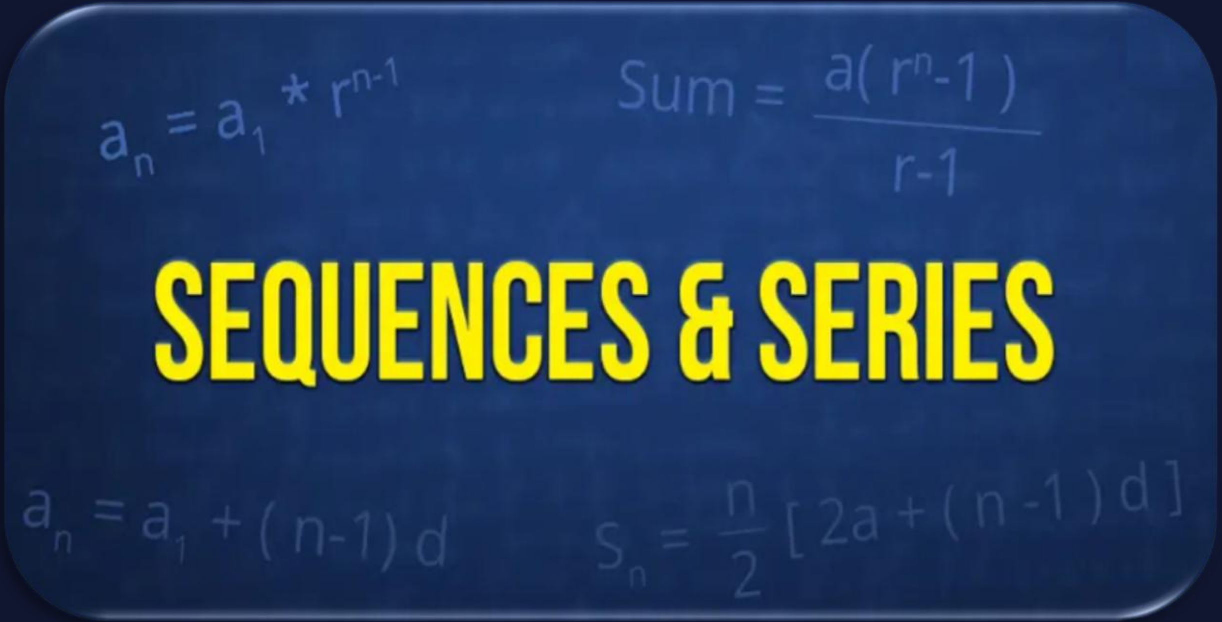
GRADE 12-Maths

UNIT 1

Unit Outcomes



Sequence & Series

Unit Outcomes

By the end of this unit, you will be able to:

- Understand sequence and series.
- Compute terms of a sequence from a given rule.
- Use given terms to develop a formula that represent the sequence.
- Identify different types of sequences and series.
- Compute the partial and infinite sum of some sequences.
- Apply your understanding the knowledge of sequences and series to real-life problems.

Arithmetic sequences:

Learning outcomes:

- Define an arithmetic sequence
- Create an arithmetic sequence
- Identify an arithmetic sequence from a given sequence and its general term
- Determine the general term of an arithmetic sequence
- Calculate the arithmetic mean of terms in an arithmetic sequence

GEOMETRIC SEQUENCE

Learning outcomes:

- Define Geometric sequence
- Create Geometric sequence
- Identify Geometric sequence from a given sequence
- Determine the general term of Geometric sequence
- Calculate the Geometric mean of terms in a Geometric sequence

1.3 The sigma notation and partial sums

Learning outcomes:

- Define a series
- Represent a series using sigma ($\sum$) notation
- Expand a series written in sigma notation into standard form
- Compute the n^{th} partial sum of a given series
- Evaluate the value of a series given in sigma notation

Sum of arithmetic and geometric progression

Learning outcomes:

- Derive and use the formula for the sum of a finite arithmetic series and finite geometric series

GRADE 12-Maths

UNIT 1 - Part 5

Infinite Series

If $|r| < 1$, the infinite geometric series

CONVERGES!

$$\sum_{n=1}^{\infty} a_1(r)^{n-1} = \boxed{\frac{a_1}{1-r}}$$

SUM

Grade 12 Maths Unit 1: Sequence and Series

Main content of the lesson	Natural Science [UEE 2003-2016]
1.4 Infinite Series	11

1.4 INFINITE SERIES

By the end of this unit, you will be able to:

- ***Define Infinite Series***
  - Identify the difference between finite and infinite series
  - Write an infinite series using sigma notation
- ***Understand Convergence and Divergence***
  - Explain what it means for an infinite series to converge or diverge
  - Determine the convergence of a geometric series
  - Apply the nth-term test for divergence
- ***Calculate Sums of Convergent Series***
- Use formulas to find the sum of an infinite geometric series
- ***Show how recurring decimals converge***
  -

1.5 APPLICATIONS OF SEQUENCE AND SERIES IN DAILY LIFE

By the end of this unit, you will be able to:

- ***discuss the applications of arithmetic and geometric progressions and series in science and technology and daily life.***

Thank you!!!!